

Axisymmetric Brill-wave collapse in AthenaK: progress, gauge controls, and outer-boundary test

Numerical investigation report

15 August 2026

1 Executive status

The calculation uses the Figure-3 Brill datum with amplitude $A = -0.047$ and independently measured ADM mass $M_{\text{ADM}} = 2.660301967997158$. IrisK’s two-dimensional single-domain spectral representation is interpolated directly inside AthenaK; the lapse is initialized as $\alpha = \psi^{-2}$. The evolution is the axisymmetric Cartoon Z4c system at nominal N128, sixth spatial order, RK4, and `dchi_max=0.02`. None of the results in this report is a completed Figure-3 reproduction or production qualification.

The strongest completed evolution remains the shock-avoiding lapse with zero shift: it covered the full paper proper-time interval before a substantially later failure. Vanilla advective $1 + \log$ plus a fixed- η Gamma driver also reached $t = 20M$. The scale-invariant telegrapher family continues to develop late curvature and constraint growth. The experiments below isolate constraint damping, KO dissipation, shift choice, AMR depth, and boundary distance without enabling a χ floor or weakening the strict-positive prolongation gates.

2 Initial-data port and early diagnostics

The original M0-wiring comparison revealed patch-like structure when the single-domain spectral datum was routed through the older domain machinery. The accepted path instead evaluates the authenticated two-dimensional IrisK coefficients directly on AthenaK cell centers. Initial evolution-variable and constraint plots were produced from AthenaK’s imported fields, rather than from an external IrisK rendering. The initial constraints are finite; the large late-time constraint features discussed here arise during evolution.

No apparent horizon finder is enabled in this Brill diagnostic series. A failure to report a horizon is therefore not evidence that a horizon is absent.

3 Gauge survey before the present five controls

The first $\max|K|$ telegrapher plus $\max|K|$ Gamma-driver run failed at $t \simeq 10.163M$. Setting Z4c $\kappa_1 = \kappa_2 = 0$ did not extend that lifetime and made the terminal constraints much larger. Replacing all $\max|K|$ coefficient normalizations by fixed coefficients delayed failure to $t \simeq 16.52M$, while changing only the shift damping from $2\max|K|$ to fixed $\eta = 2$ delayed it to $t \simeq 16.74M$. Thus runaway shift damping was a major stiffness source, but not the only instability.

Three field-local telegrapher scales— $|K|$, $\sqrt{K_{ij}K^{ij}}$, and $\sqrt{\gamma^{ij}\partial_i\chi\partial_j\chi}$ —all failed earlier than the domain-maximum control. Their spatial profiles are reproduced in Fig. 1. Scaling τ and κ together from (1,1) to (2,2) reduced the discrete damping stiffness while keeping wave speed unchanged, but both level-20 evolutions failed at central proper time $\tau \simeq 10.38M$.

The shock-avoiding/zero-shift and vanilla-1 + log/Gamma-driver comparisons are retained in Figs. 2–3. Their paper curves are rendered-vector extractions because raw numerical Figure-3 data were not published.

case	outer $R = Z$	shift	KO	t_{last}	τ_{last}	α_{axis}	L_{max}	AH	terminal
1	16	fixed $\eta = 2$	0.02	16.722	10.365	0.64398	20	no	strict positive- χ gate
2	16	fixed $\eta = 2$	0.5	16.714	10.372	0.64496	20	no	exit 1
3	16	zero	0.5	7.4926	4.5815	0.5979	13	no	exit 1
4	64	fixed $\eta = 2$	0.5	16.714	10.372	0.64496	20	no	strict positive- χ gate
5	64	zero	0.5	7.4927	4.5816	0.5979	20	no	strict positive- χ gate

Table 1: The five requested controls. Last values are the last fully finite history rows, not post-failure extrapolations. The AH column is recorded telemetry only; the horizon finder was disabled for this diagnostic series.

4 Five zero-damping KO/shift/domain controls

All five cases use the pre-collapsed telegrapher lapse, $(\tau, \kappa) = (1, 1)$, $\max|K|$ scaling, and exactly zero Z4c constraint damping. Cases 1–3 use $0 \leq \rho \leq 16$, $|z| \leq 16$. Cases 4–5 repeat the later KO=0.5 cases on $0 \leq \rho \leq 64$, $|z| \leq 64$. The base spacing remains 0.25 because the root grids grow from 64×128 to 256×512 ; the larger-box test does not trade boundary distance for resolution.

The original KO=0.5 cases approach the 8,192-block operational guard. At the user’s direction, the enlarged-domain cases run sequentially on one A100 in the shared-interactive queue. The single-rank and global guards are therefore both 16,384 blocks. That capacity change is not an AMR-criterion change: `dchi_max=0.02` and maximum physical level 20 remain fixed. It is reported explicitly because an earlier capacity or GPU-memory stop is a scaling result, not a boundary-condition diagnosis.

The first single-GPU attempt used an A100-SXM4-40GB. Both cases imported the IrisK data and passed the finite initial constraint conversion, but Kokkos then failed before evolution while allocating the 4.883-GiB load-balance send buffer. Because the 16,384-block single-rank contract reserves paired send and receive buffers, reducing this capacity would reintroduce the known AMR block guard rather than test the boundary. The strict successor therefore kept one GPU and every science parameter unchanged while requiring an A100-SXM4-80GB. The five-case plots use only that successor’s evolved data; the 40-GiB attempt is retained as operational evidence, not counted as a sixth science curve.

The 80-GiB successor ran the two enlarged-domain cases strictly one at a time. The fixed-shift case reached its numerical endpoint in the first numbered step. The following zero-shift step was still evolving normally when its per-step wall-time expired. A fresh one-case successor therefore resumes only that zero-shift case from its latest complete authenticated restart, on one shared-interactive A100-SXM4-80GB. The pre-restart and continuation histories are joined at the restart time and plotted as one physical case; the restart is not a sixth gauge/domain configuration.

The timestep is dynamic through the CFL/AMR controller, not through an explicit $\max|K|$ stiffness estimate. For this Z4c implementation the candidate timestep is the MPI-global minimum represented cell width times `cfl_number=0.15`. With base spacing 0.25, level L therefore gives $\Delta t = 0.0375/2^L$: levels 8, 9, 10, and 20 correspond respectively to 1.46484×10^{-4} , 7.32422×10^{-5} , 3.66211×10^{-5} , and 3.57628×10^{-8} . The controller does not currently insert $\max|K|$, lapse, shift, τ , κ , or a measured gauge speed into that Z4c estimate. Hence severe wall-time growth in these runs is directly tied to refinement depth, even when the gauge coefficients are held fixed.

Horizon status. No finite history row in any of the five cases reported an apparent horizon.

5 Interpretation and remaining uncertainty

The enlarged-domain runs are designed to answer a narrow causal question: whether late artifacts move materially when the outflow boundary is causally farther away. Agreement before the original light-crossing

time disfavors a direct outer-boundary explanation; divergence does not by itself prove a boundary reflection because the AMR tree can differ. The pairwise plots and terminal inventories must therefore be interpreted together.

For the fixed-shift $KO=0.5$ pair, interpolation at matched coordinate times shows that the axis lapse agrees to better than 8×10^{-8} and the axis curvature to about 1.2×10^{-6} through $t = 16$. The terminal times are likewise essentially identical: 16.7140508 on the original box and 16.7140514 on the enlarged box. The global constraint norms differ more because their volume integrals cover different domains, but the central gauge and curvature histories rule out the original outer boundary as the driver of this fixed-shift failure. The zero-shift pair agrees even more closely through the original box’s terminal time, $t = 7.4925751$. At that matched time the relative differences in the global C , H , and M norms are respectively 5.3×10^{-6} , 1.1×10^{-5} , and 2.6×10^{-5} ; $\max|K|$ differs by 1.8×10^{-5} and the global maximum curvature by 1.2×10^{-6} . The axis lapse differs by only 1.0×10^{-12} , central proper time by 1.3×10^{-14} , and axis curvature by 7.8×10^{-11} relatively. The small Z norm is more volume-sensitive and differs by 17%, but this does not accompany a central trajectory difference. The enlarged case then passes the original 8,192-block stop, reaches the configured maximum physical level 20 and 13,580 blocks, and continues to display the same catastrophic constraint/curvature growth. It finally stops at $t = 7.4927181$ when the unchanged strict gate rejects 2,346 nonpositive or nonfinite parent stencils for χ . The original boundary is therefore not the cause of the zero-shift instability; its lower capacity only selected the earlier terminal event. The full terminal comparison is reported in Table 1 and Fig. 7.

The current evidence does not support curing the telegrapher evolution by arbitrarily raising KO dissipation, enabling a χ floor, or selectively relaxing a strict gate. The most promising algorithmic direction remains a smoother scale-invariant gauge scale than a domain maximum, together with explicit history telemetry and restart-authoritative filtering. Any such change needs a prospective matched comparison rather than tuning to one terminal time.

Taken together, these controls disfavor three simple explanations as the sole cause. The fixed and zero-shift choices both fail, although at different times; raising KO dissipation from 0.02 to 0.5 does not remove the late growth; and moving the outer boundary from 16 to 64 leaves the central trajectories and terminal times essentially unchanged. This does not prove that gauge or dissipation is irrelevant, but it means that another blind parameter sweep is not the most discriminating next experiment.

The cleanest prospective resolution test is to increase the cell count inside each MeshBlock while retaining the logical root-block layout, physical domain, maximum AMR level, $d\chi$ threshold, gauge, and KO coefficient. This raises the resolution at every represented level without deliberately changing the refinement policy. Dynamic thresholding can nevertheless produce a different tree; therefore the run must archive physical refinement coverage and block decisions, or use a prescribed/replayed refinement mask if literal tree identity is required. Pre-terminal convergence of the constraints and a systematic delay of the terminal event with decreasing spacing would favor under-resolution or a numerical instability. A terminal event at nearly the same physical time and location would instead favor a formulation/gauge limitation. A separate, one-factor-at-a-time matrix should vary the initial base resolution and $d\chi_{\max}$ to measure AMR-criterion sensitivity; these changes should not be combined in the first resolution comparison. None of these proposed runs was executed for this report.

6 Reproducibility boundaries

The final artifact binds the selected original-domain ledger, enlarged-domain initial and continuation jobs, terminal manifests, settled accounting, comparison JSON, histories, logs, rank bindings, plotting inputs, and derived files by SHA-256. Raw binary/restart hierarchies remain at their immutable Perlmutter roots and are not silently reinterpreted. All numerical failures are retained. This report makes no horizon, convergence, or complete Figure-3 reproduction claim.

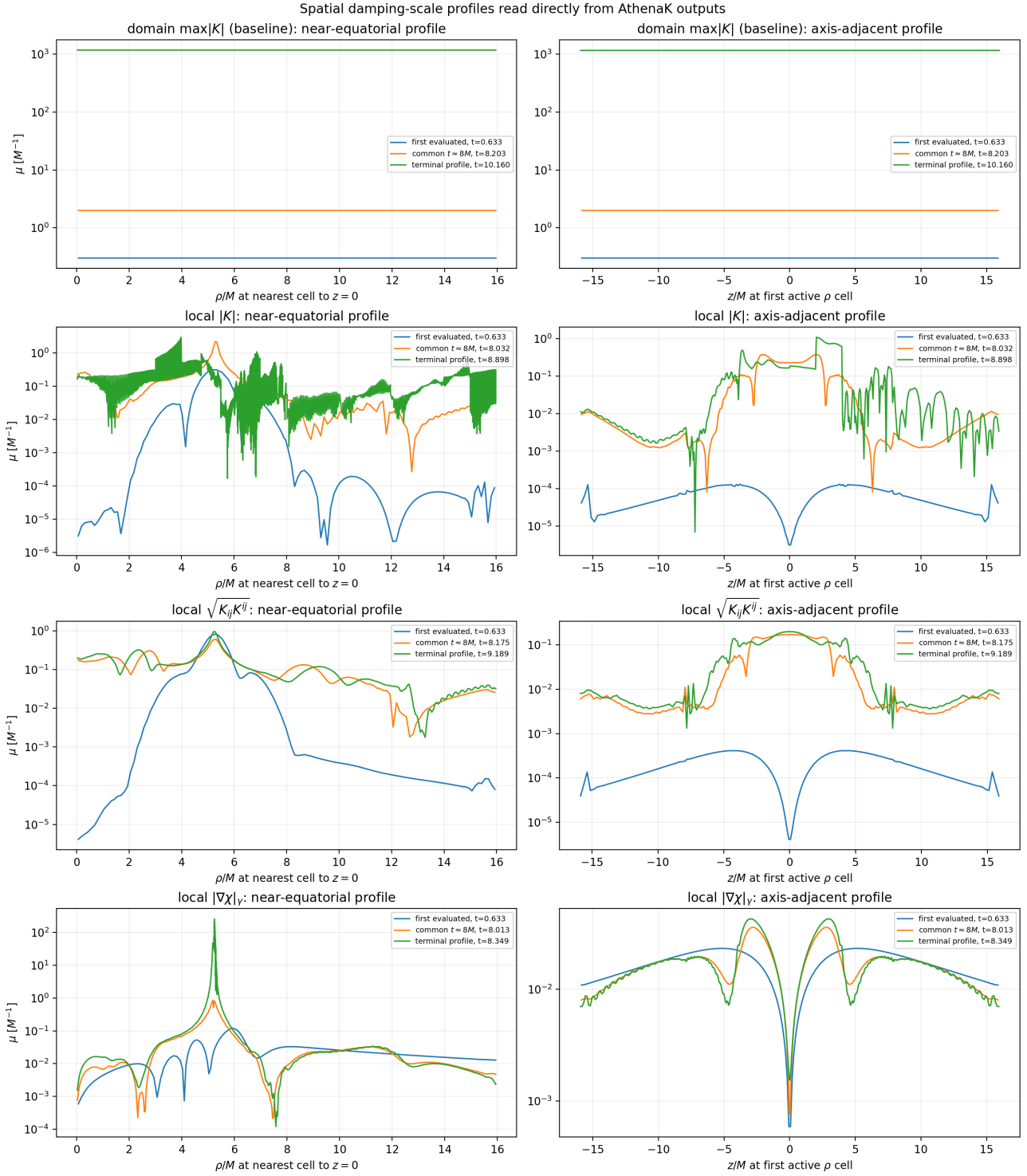


Figure 1: Previously authenticated AthenaK spatial profiles for the tested telegrapher damping scales. They are evolved AthenaK fields, not IrisK interpolation plots.

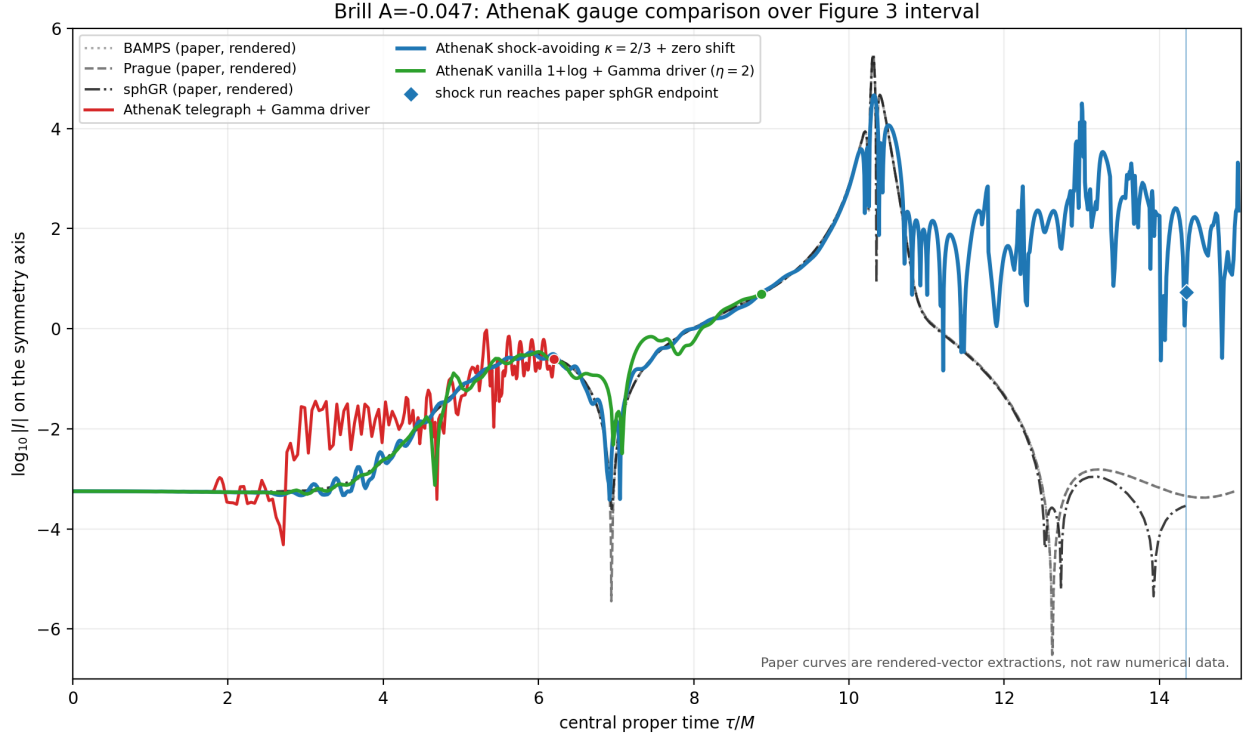


Figure 2: Earlier gauge-family overlay against the rendered Figure-3 curves.

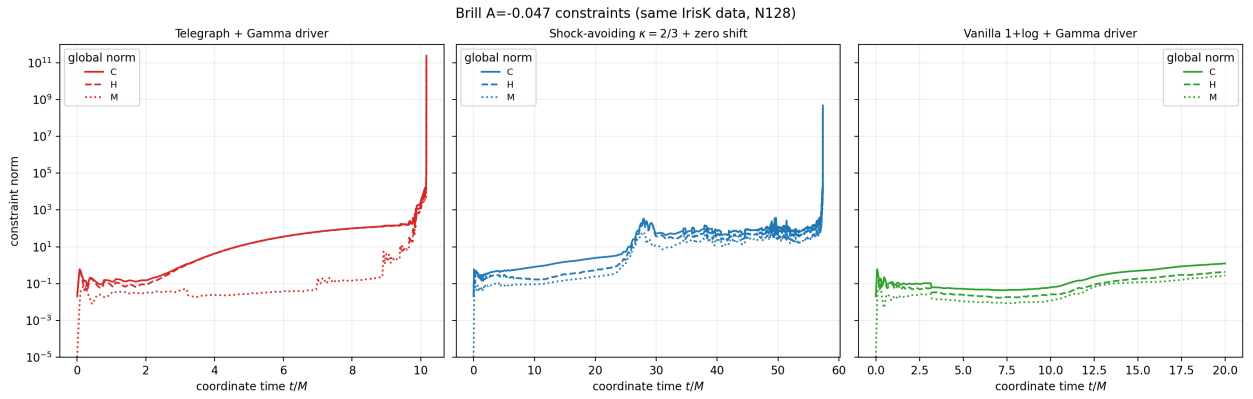


Figure 3: Earlier constraint comparison for telegrapher, shock-avoiding, and vanilla gauge families.

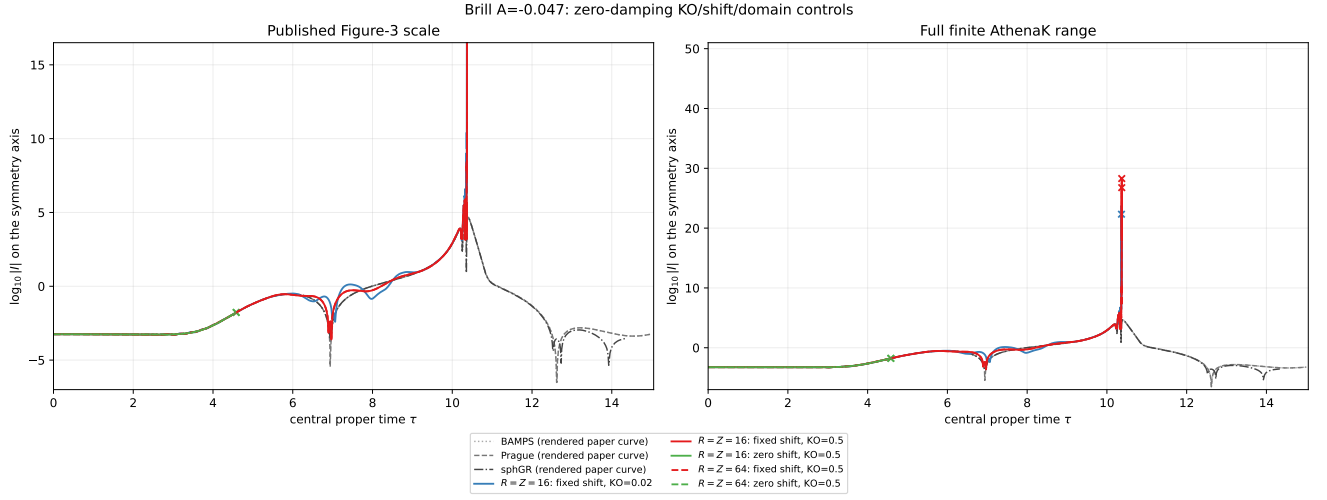


Figure 4: All five requested controls overlaid on the rendered published Figure-3 curves. Crosses mark each last finite AthenaK row.

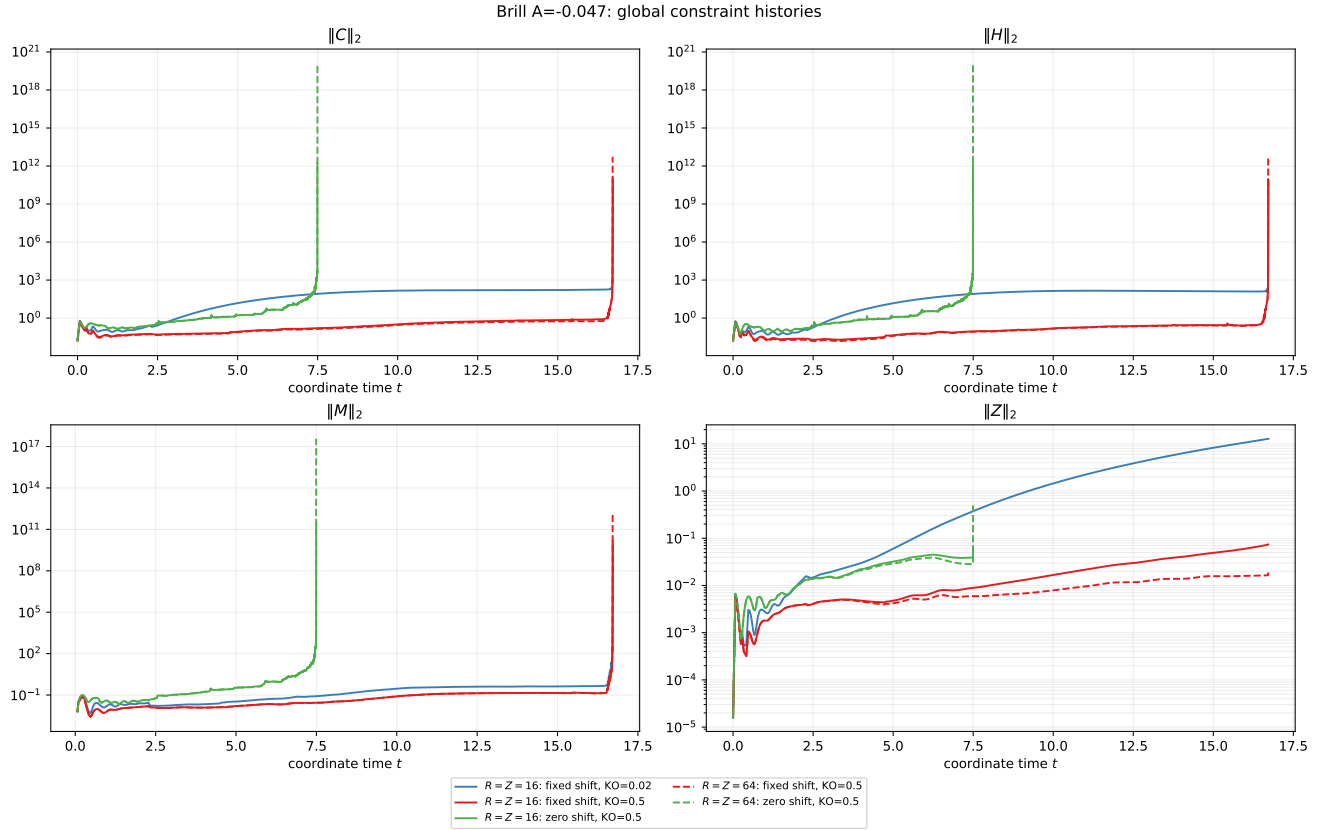


Figure 5: Global Z4c constraint histories for all five controls.

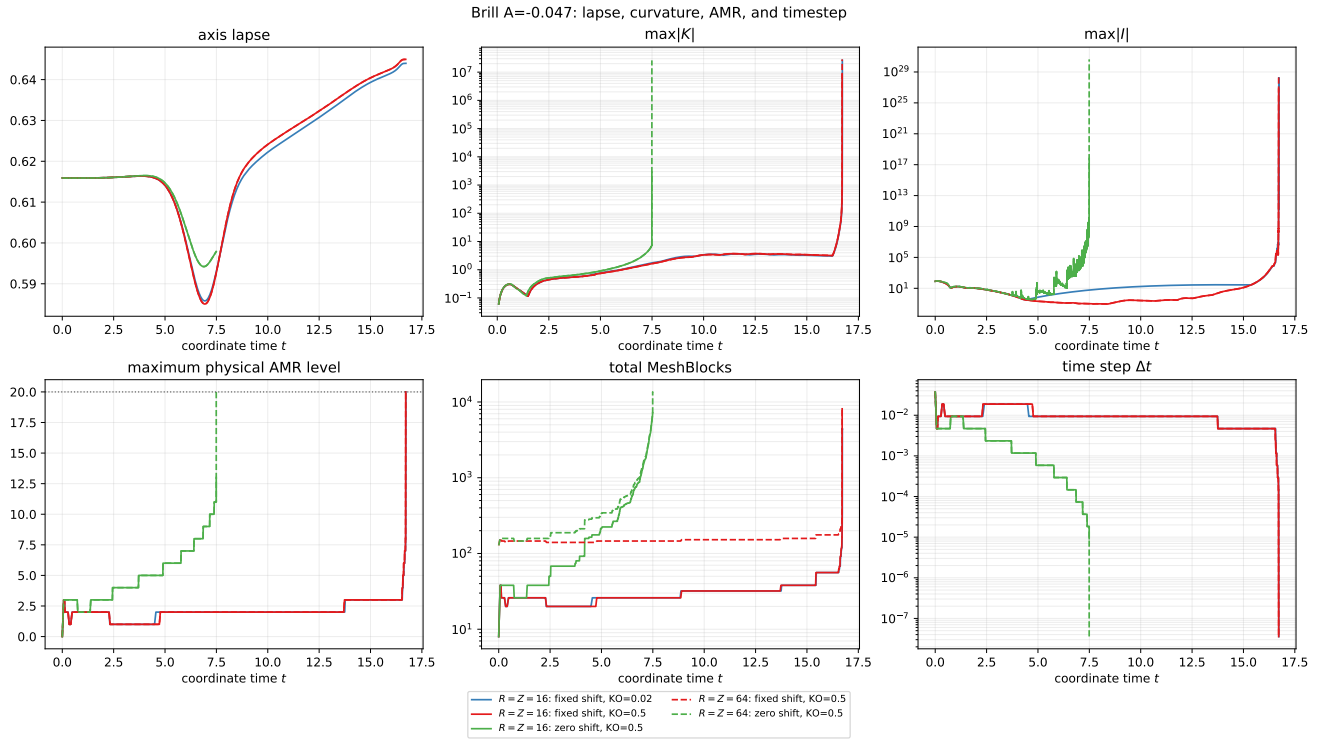


Figure 6: Axis lapse, curvature, AMR inventory, and timestep diagnostics.

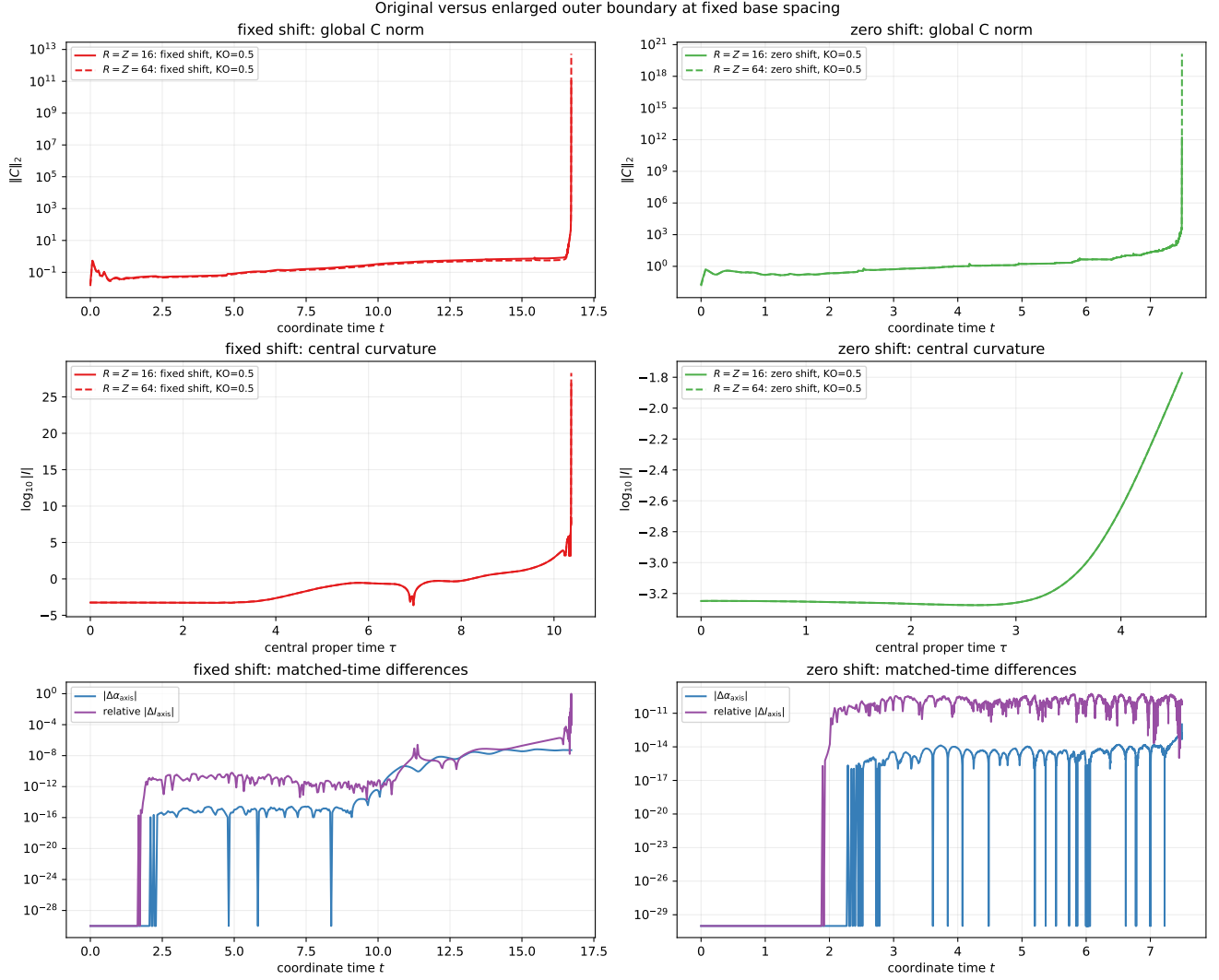


Figure 7: Pairwise original-domain/enlarged-domain comparisons for the two $KO=0.5$ shift choices. Solid and dashed lines differ in boundary distance while retaining the same base spacing.